ARALert Deployment Tech Stack & Feasibility Report

Zero-Cost Cloud Architecture, Resource Quotas, Scale Modeling & Backup Protocol

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1. DEPLOYMENT ARCHITECTURE & ROLES

FRONTEND WEB APP

Host: Vercel (Hobby Tier)

- React 19 + TypeScript + Vite
- Tailwind CSS & Canvas QR
- Global Edge CDN & Auto-SSL

Cost: $0.00 / month

BACKEND API SERVICE

Host: Render (Free Tier)

- Python 3.13 + Django REST
- Gunicorn + WSGI / JWT Auth
- 750 Free Instance Hours/Mo

Cost: $0.00 / month

RELATIONAL DATABASE

Host: Supabase (Free Tier)

- Managed PostgreSQL 15+
- Connection Pooling (Supavisor)
- 500 MB Storage / 5 GB Egress

Cost: $0.00 / month

2. ZERO-COST FREE-TIER QUOTA COMPLIANCE MATRIX (OCTOBER 2026)

Infrastructure Resource	Free Tier Limit	ARALert Usage (300 Learners)	Quota Used	Monthly Cost
Vercel Bandwidth (Data Transfer)	100 GB / month	~2.0 GB / month	2.00%	PHP 0.00 ($0)
Vercel Custom Domains & SSL	Unlimited (Free SSL)	1 Domain (aralert.vercel.app)	Included	PHP 0.00 ($0)
Render Web Instance Hours	750 Hours / month	720 Hours (Continuous 30 days)	96.00%	PHP 0.00 ($0)
Render Instance Memory / CPU	512 MB RAM / 0.1 CPU	~140 MB RAM (Gunicorn/Django)	27.34%	PHP 0.00 ($0)
Supabase PostgreSQL Storage	500 MB (0.5 GB)	~12.87 MB / Semester	2.57%	PHP 0.00 ($0)
Supabase Bandwidth (Egress)	5 GB / month	~200 MB / month	4.00%	PHP 0.00 ($0)
Supabase Connection Pooling	Included via Supavisor	Direct & Pooled Connections	Included	PHP 0.00 ($0)
Automated App Backup Storage	15 GB (Google Drive / Local)	~90 MB / Semester (.json.gz)	0.60%	PHP 0.00 ($0)
NET TOTAL OPERATIONAL DEPLOYMENT COST				PHP 0.00 / MONTH ($0.00)

3. KEY FEASIBILITY TAKEAWAYS FOR FACULTY & STAKEHOLDERS

1. Complete Zero-Cost Viability

All core services (frontend, API, database, backups) fit inside standard free-tier quotas with zero risk of incurring fees or hitting paywalls during the 1-week pilot testing and regular academic semester.

2. Highly Optimized Data Footprint

Because ARALert stores structured alphanumeric data (QR hashes, attendance records, Phil-IRI scores) rather than heavy media, total semester data occupies only ~12.87 MB (2.57% of Supabase 500MB).

3. Supabase Free Tier Backup Strategy

While Supabase provides managed PostgreSQL, automated dashboard backups are a Pro tier feature ($25/mo). Our Django ORM-level automated backup routine (JSON dumps) provides complete data protection at $0 cost.

4. High Availability During School Hours

Render's 750 free instance hours fully cover standard school operating schedules (7:00 AM to 5:00 PM), ensuring responsive performance during daily live tutoring sessions.

4. MATHEMATICAL DATA VOLUME & SCALE PROJECTIONS

System Table / Entity	1-Week Pilot (4 Days)	1 Semester (20 Weeks)	1 School Year (40 Wks)	Postgres Size
User Accounts & Profiles	320 rows	320 rows	350 rows	~140 KB
Student Enrollments & Tutor Assignments	600 rows	600 rows	1,200 rows	~180 KB
Attendance Sessions (QR Portals)	60 sessions	1,200 sessions	2,400 sessions	~480 KB
Learner Attendance Logs	1,200 scans	24,000 scans	48,000 scans	~5.76 MB
Milestone Assessment Grades	600 scores	4,500 scores	9,000 scores	~1.62 MB
Academic Risk Evaluations (30/50/20)	300 records	6,000 records	12,000 records	~1.80 MB
Security Audit Logs & Edit Trails	2,000 logs	15,000 logs	30,000 logs	~7.50 MB
JWT Blacklisted Tokens & Auth Sessions	500 tokens	1,000 tokens	2,000 tokens	~400 KB

TOTAL ESTIMATED DATABASE FOOTPRINT (WITH INDEXES & WAL):
1-Week Pilot: ~1.26 MB (0.25% DB Quota) | 1 Semester: ~12.87 MB (2.57%) | 1 School Year: ~25.74 MB (5.15%)

5. DATABASE LONGEVITY ANALYSIS

Under Supabase's 500 MB free allocation, ARALert's projected annual growth rate of 25.74 MB per school year provides approximately 19.4 consecutive years of continuous operational logging before reaching capacity. Furthermore, completed school years can be archived into historical JSON files, enabling perpetual zero-cost operation indefinitely.

6. NETWORK BANDWIDTH & EGRESS CONSUMPTION

Vercel Frontend Bandwidth
Free Quota: 100 GB / month
- Single-Page App initial load: ~1.2 MB
- Daily cached visits across all users: ~65 MB
- Monthly Total: ~1.95 GB / 100 GB (1.95% Used)

Supabase Database Egress
Free Quota: 5 GB / month
- QR scan payload response: ~0.8 KB per scan
- Dashboard telemetry queries: ~5.5 MB / day
- Monthly Total: ~195 MB / 5,000 MB (3.90% Used)

7. 4-TIER DISASTER RECOVERY & ZERO-COST BACKUP PROTOCOL

Because Supabase Free Tier reserves automated dashboard snapshots for Pro plans ($25/mo), ARALert utilizes an automated, application-level 4-tier backup strategy to guarantee zero data loss and compliance with RA 10173.

Tier 1: Cloud Real-Time Transaction Safety

Schedule: Continuous | Tool: Supabase WAL Logging

PostgreSQL Write-Ahead Logging (WAL) ensures every committed QR attendance scan and milestone grade is written to disk instantly. Even during sudden power cuts, incomplete transactions rollback safely with zero table corruption.

Tier 2: Midday Assessment Snapshot

Schedule: 12:00 PM Daily | Tool: Django ORM dumpdata

Captures all morning session attendance logs and preliminary assessment entries. Stored as a compressed .json.gz file (~190 KB), ensuring that afternoon activities cannot jeopardize morning progress.

Tier 3: End-of-Day Master Snapshot

Schedule: 5:00 PM Daily | Tool: Django ORM dumpdata

Locks in the entire day's operations, including tutor qualitative remarks, student milestone evaluations, and faculty DTR records. Backups are automatically archived to the local administrator PC and synced to Google Drive.

Tier 4: Event-Driven Pre-Update Snapshot

Schedule: On-Demand | Tool: Pre-Deployment Hook

Executed immediately prior to applying any system updates, Git pulls, or Django database migrations. If any error or broken migration occurs, the system triggers an instantaneous rollback (<5 minutes).

8. OPERATIONAL BEST PRACTICES FOR PILOT TESTING WEEK

Morning Warm-Up Routine (Render Sleep Policy)

Because Render's free tier spins down after 15 minutes of inactivity, the lead tutor or administrator should open the ARALert landing page at 7:45 AM. The initial 45-second cold start will spin up the server, ensuring instant QR scanning when students arrive at 8:00 AM.

Backup Retention & Storage Management

Store compressed JSON backups in a dedicated folder (/backups/) with 30-day rotation. Total backup storage for the entire school semester will remain under 90 MB, easily fitting inside a standard free 15 GB Google Drive or USB drive.

Offline Failover & Emergency Ledger

Tutors should keep the offline PDF Attendance & Milestone Scoring Ledger printed (from ARALert's export module) in the classroom as a secondary paper failover in case of rural Masbate telecommunication blackouts.

Semester Archive Export & DepEd Registers

At the end of each academic semester, the system administrator can run a full database export and download all student accomplishment registers to finalize DepEd Form 137 records.

SYSTEM READINESS VERIFICATION: APPROVED FOR PILOT DEPLOYMENT

Compliant with ISO/IEC 25010 Quality Standards, RA 10173 Data Privacy Act, and DepEd ARAL Program Mandates.